

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO2:** Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Industry Problem Task

Duration: 2.5hrs

1. Illustrate how Reinforcement Learning is used in a streaming platform for content recommendation. Discuss the impact of delayed rewards, user feedback, and changing preferences on learning performance.

Objective:

The objective is to develop an intelligent recommendation system that suggests movies, TV shows, or videos based on each user's preferences and viewing history. The system should continuously learn from user interactions and improve recommendations to maximize user satisfaction, watch time, and platform engagement.

Why Reinforcement Learning (RL) is Suitable?

Reinforcement Learning is suitable because:

- Users' interests change over time.
- The system learns from continuous user feedback.
- RL balances recommending popular content (exploitation) with discovering new content (exploration).
- It adapts automatically without requiring frequent retraining.
- It optimizes long-term user engagement rather than only immediate clicks.

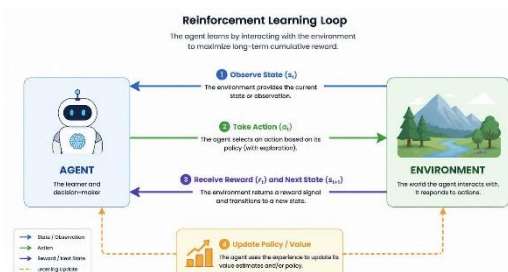
Agent and Environment

Agent

The recommendation engine (AI system).

Environment

The streaming platform and its users, where the agent interacts by recommending content and observing user responses.



State

The **state** represents the current information about a user.

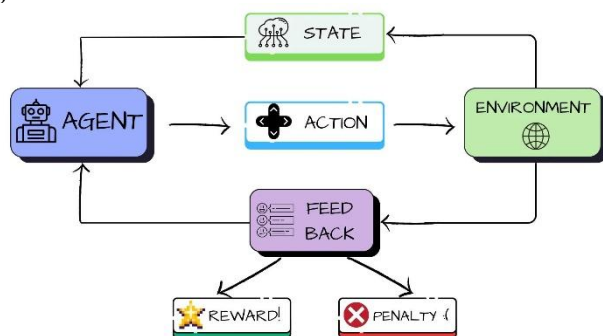
Example State:

- Viewing history
- Previously watched genres
- Search history
- Time of day
- Device used
- Watch duration
- User ratings
- Subscription type
- Recently skipped content

Example:

State S =

```
{
Comedy,
Watched 5 action movies,
Night (9 PM),
Mobile device,
Likes Sci-Fi
}
```



Actions

The **action** is recommending content.

Examples:

- Recommend Movie A
- Recommend Movie B
- Recommend TV Series
- Recommend Documentary
- Recommend Trending Content
- Recommend New Release

Reward Function

Positive Rewards (+)

User Action	Reward
Clicks recommendation	+2
Watches completely	+10
Gives Like	+5
Rates 5 stars	+8

Watches another recommendation	+6
Subscription renewal	+20

Negative Rewards (-)

User Action	Reward
Ignores recommendation	-2
Skips after few seconds	-5
Gives dislike	-6
Stops watching	-8
Cancels subscription	-20

Policy Used by the Agent

A **policy** decides which recommendation to show.

Common policy:

ϵ -Greedy Policy

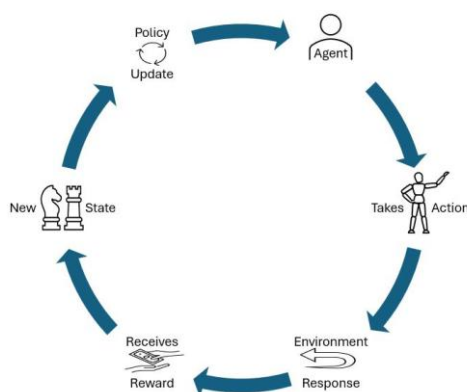
- **Exploration:** Recommend new or unseen content with probability ϵ .
- **Exploitation:** Recommend content known to interest the user with probability $(1-\epsilon)$.

Example:

90% → Recommend favorite genre

10% → Recommend new genre

This prevents the recommendation system from becoming repetitive.



Learning Process

Step 1

Observe user's current state.

↓

Step 2

Recommend content.



Step 3

User interacts with recommendation.



Step 4

Receive reward.



Step 5

Update Q-value.

Q-learning update equation:

$$Q(s, a) = Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Where:

- **α (Learning Rate):** Controls how quickly learning occurs.
- **γ (Discount Factor):** Determines importance of future rewards.
- **r :** Immediate reward.
- **s' :** Next state.

The process repeats continuously.

Continuous Learning

The recommendation system improves over time because:

- Learns new user interests.
- Detects seasonal trends.
- Adapts when users change preferences.
- Improves recommendation accuracy.
- Reduces irrelevant recommendations.
- Increases watch time.
- Improves customer retention.

Example:

Initially:

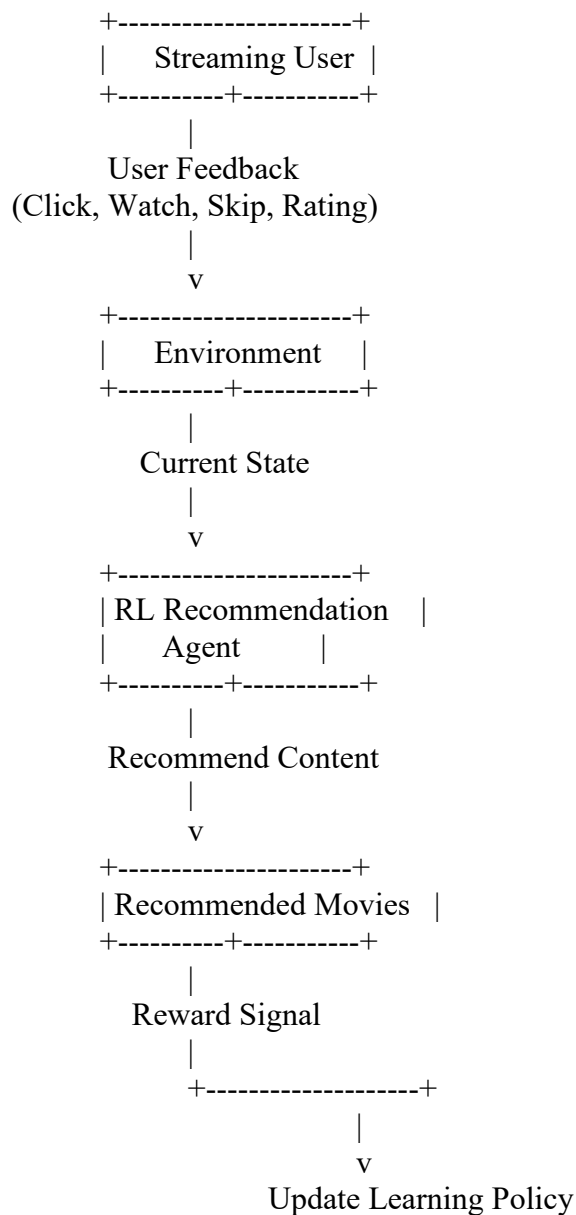
User watches Action movies.

After several weeks:

User starts watching Comedy.

RL automatically recommends more Comedy movies.

RL Architecture (Block Diagram)



Real-World Industry Example

Netflix

Netflix uses AI-based recommendation systems to suggest movies and TV shows based on:

- Viewing history
- Search behavior
- Watch duration

- Ratings
- Time of watching
- Similar users' preferences

The recommendation engine continuously learns from user interactions, improving personalization and increasing watch time.

Other examples:

- YouTube
- Amazon Prime Video
- Disney+
- Spotify (music recommendation)

Delayed Rewards

Delayed rewards occur when the benefit of a recommendation is observed later rather than immediately.

Example:

Day 1:

Recommend "Stranger Things"

User ignores it.

Day 5:

User searches and watches the same show.

Although there was no immediate reward, the recommendation influenced future behavior. RL captures these delayed rewards using the discount factor (γ), allowing the agent to value long-term outcomes.

User Feedback

The recommendation system learns from both explicit and implicit feedback.

Explicit Feedback

- Ratings
- Likes
- Dislikes
- Reviews

Implicit Feedback

- Watch time
- Clicks
- Skips
- Replays

- Search history
- Time spent browsing

More feedback leads to better recommendations.

Changing Preferences

User preferences evolve over time.

Example:

Month	Preferred Genre
January	Action
March	Romance
June	Comedy
September	Documentaries

Advantages of Reinforcement Learning

- Personalized recommendations
- Continuous learning
- Adapts to changing user interests
- Handles delayed rewards effectively
- Improves user engagement
- Increases watch time
- Boosts customer satisfaction
- Reduces irrelevant recommendations
- Supports long-term optimization
- Improves subscription retention

Expected Outcomes

- Higher recommendation accuracy
- Increased click-through rate (CTR)
- Longer viewing sessions
- Better user experience
- Improved customer loyalty
- Increased subscription renewals
- Reduced user churn
- Higher platform revenue
- Adaptive and scalable recommendation system
- Continuous improvement through real-time learning

Conclusion

Reinforcement Learning enables streaming platforms to deliver intelligent, personalized content recommendations by learning from user interactions, delayed rewards, and evolving preferences. Through continuous feedback and exploration–exploitation strategies, the system adapts over time, resulting in higher user engagement, improved recommendation accuracy, increased watch time, and stronger customer retention. This makes RL a highly effective approach for modern streaming services such as Netflix, YouTube, Disney+, and Amazon Prime Video.

2.Evaluate the effectiveness of Reinforcement Learning in predictive maintenance compared to traditional rule-based approaches. Discuss risks and reliability concerns in industrial environments.

1. Introduction

Predictive maintenance aims to detect equipment faults before failure occurs by analyzing machine conditions and scheduling maintenance at the right time. Traditional rule-based systems rely on predefined rules and fixed thresholds, whereas Reinforcement Learning (RL) continuously learns from machine data and maintenance outcomes to make better maintenance decisions over time.

2. Comparison: Reinforcement Learning vs. Rule-Based Approach

Feature	Reinforcement Learning (RL)	Traditional Rule-Based Approach
Learning Ability	Learns continuously from data and experience	Uses fixed rules defined by experts
Adaptability	Adapts to changing machine conditions	Difficult to adapt when conditions change
Decision Making	Optimizes long-term maintenance strategies	Makes decisions based on predefined conditions
Accuracy	Improves over time with more data	Accuracy remains constant unless rules are updated
Maintenance Cost	Reduces unnecessary maintenance	May perform maintenance too early or too late
Scalability	Suitable for complex industrial systems	Difficult to manage many rules for large systems
Human Intervention	Minimal after deployment	Frequent manual updates required

3. Effectiveness of Reinforcement Learning

Reinforcement Learning offers several advantages in predictive maintenance:

- Continuously learns from sensor data and machine performance.
- Detects faults earlier than fixed-rule systems.
- Reduces unexpected equipment failures.
- Optimizes maintenance schedules based on machine health.
- Minimizes maintenance costs by avoiding unnecessary servicing.
- Increases equipment lifespan.
- Improves production efficiency and system availability.
- Adapts automatically to new operating conditions.

4. Risks in Industrial Environments

Although RL is powerful, it also introduces certain risks:

- **Safety Risks:** Incorrect decisions may lead to equipment damage or accidents.
- **Data Quality Issues:** Poor or inaccurate sensor data can reduce learning performance.
- **Exploration Risk:** During learning, the agent may choose unsafe maintenance actions.
- **High Computational Requirements:** Training RL models requires significant computational resources.
- **Delayed Learning:** Large amounts of operational data are needed before achieving reliable performance.

5. Reliability Concerns

Industries require highly reliable maintenance systems because failures can result in significant financial losses or safety hazards.

Key reliability concerns include:

- Consistent decision-making under different operating conditions.
- Robustness to sensor failures or missing data.
- Ability to explain maintenance decisions to engineers.
- Stability when machinery or operating environments change.
- Compliance with industrial safety standards.

To improve reliability, RL systems are often combined with:

- Human expert supervision.
- Safety constraints.
- Predictive analytics.
- Regular model validation and testing.

6. Real-World Example

In a manufacturing plant, sensors monitor the temperature, vibration, and pressure of industrial motors.

- A **rule-based system** schedules maintenance whenever vibration exceeds a fixed threshold (e.g., 50 mm/s), regardless of other factors.
- An **RL-based system** analyzes multiple sensor readings, historical failures, and maintenance outcomes to determine the optimal maintenance time, reducing downtime and maintenance costs while improving machine reliability.

7. Advantages of Reinforcement Learning

- Reduces unexpected equipment failures.
- Improves maintenance planning.
- Lowers operational and maintenance costs.
- Increases machine availability.
- Extends equipment lifespan.
- Continuously improves through experience.
- Adapts to changing industrial environments.

- Enhances overall production efficiency.

8. Conclusion

Reinforcement Learning is more effective than traditional rule-based approaches for predictive maintenance because it continuously learns from operational data, adapts to changing machine conditions, and optimizes maintenance decisions for long-term performance. While RL offers significant improvements in efficiency, cost reduction, and equipment reliability, it also requires high-quality data, careful safety measures, and continuous monitoring to ensure reliable operation in industrial environments. When implemented with appropriate safeguards, RL provides a smarter and more adaptive predictive maintenance solution for modern industries.

3. Analyze how Reinforcement Learning can be used to optimize transportation systems such as bus scheduling or route allocation. Define states, actions, and rewards, and discuss how real-time data improves performance

1. Introduction

Transportation systems face challenges such as traffic congestion, fluctuating passenger demand, delays, and inefficient route allocation. Reinforcement Learning (RL) enables transportation management systems to learn optimal scheduling and routing decisions by interacting with real-time traffic conditions and passenger demand. The goal is to reduce travel time, improve service quality, and maximize operational efficiency.

2. Why Reinforcement Learning?

Reinforcement Learning is suitable because it:

- Learns from real-time traffic conditions.
- Adapts to changing passenger demand.
- Optimizes long-term transportation efficiency.
- Reduces delays and waiting times.
- Improves bus utilization and fuel efficiency.
- Continuously updates its routing strategy based on feedback.

3. State

The **state** represents the current condition of the transportation system.

Example State Variables

- Current bus location
- Traffic congestion level
- Number of waiting passengers
- Weather conditions
- Time of day
- Bus occupancy
- Road conditions
- Distance to next stop

Example State:

State S =

{

Bus at Stop A,

Traffic = Heavy,

Passengers Waiting = 45,

Time = 8:30 AM,

Occupancy = 80%

}

4. Actions

The **action** is the decision taken by the RL agent.

Examples include:

- Dispatch another bus
- Change bus route
- Delay departure
- Increase bus frequency
- Reduce bus frequency
- Select shortest route
- Select least congested route
- Assign additional buses
- Skip low-demand stops (if permitted)

5. Reward Function

The reward function encourages efficient transportation decisions.

Positive Rewards (+)

Action	Reward
Bus arrives on time	+10
Reduced passenger waiting time	+8
Fuel-efficient route selected	+6
High passenger satisfaction	+10
Reduced traffic congestion	+8
Increased bus utilization	+7

Negative Rewards (-)

Action	Reward
Bus delay	-8
Traffic congestion	-6
Empty bus trip	-5
Passenger complaints	-7
Route deviation	-4
Missed schedule	-10

6. Learning Process

The RL agent follows these steps:

1. Observe the current transportation state.
2. Select the best route or schedule.
3. Operate the bus according to the selected action.
4. Receive feedback from passengers and traffic conditions.
5. Calculate the reward.
6. Update the learning policy.
7. Repeat the process continuously.

The agent gradually learns which scheduling and routing strategies provide the highest long-term rewards.

7. Real-Time Data Improves Performance

Real-time data significantly enhances the RL system by allowing immediate adaptation to changing conditions.

Sources of Real-Time Data

- GPS location of buses
- Live traffic information
- Passenger counts
- Road accidents
- Weather updates
- Smart ticketing systems
- CCTV and traffic sensors

Benefits

- Dynamic route optimization
- Faster response to traffic congestion
- Reduced travel time
- Improved schedule accuracy
- Better passenger experience
- Lower fuel consumption

8. Real-World Example

Smart City Bus System

A city's public transport authority uses Reinforcement Learning to manage bus operations.

- GPS continuously tracks bus locations.
- Traffic sensors report congestion levels.
- Passenger counts are collected at each stop.
- The RL agent analyzes this information in real time.
- It dynamically changes bus routes and adjusts bus frequency to minimize delays and waiting times.

As the system gains more experience, it continuously improves scheduling efficiency and service quality.

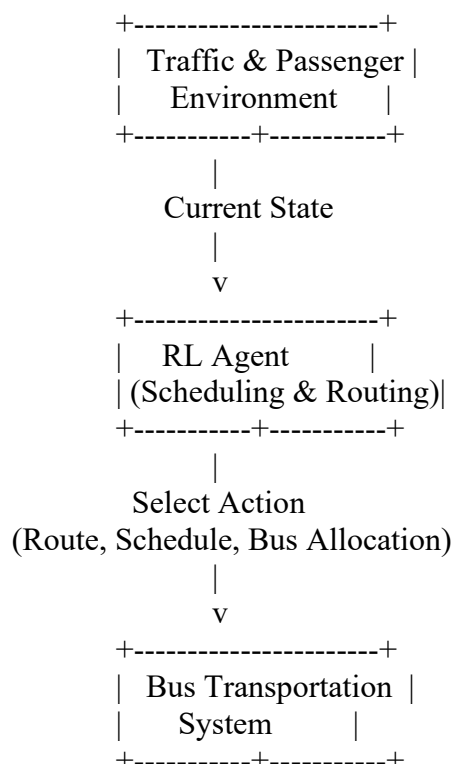
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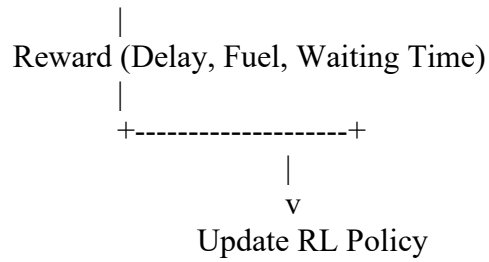
- Optimizes bus scheduling automatically
- Reduces passenger waiting time
- Adapts to changing traffic conditions
- Improves route allocation
- Reduces fuel consumption
- Lowers operational costs
- Increases passenger satisfaction
- Supports real-time decision making
- Continuously improves through experience
- Enhances overall transportation efficiency

10. Expected Outcomes

- Reduced traffic congestion
- Improved bus punctuality
- Better route utilization
- Shorter travel times
- Higher passenger satisfaction
- Increased operational efficiency
- Lower maintenance and fuel costs
- Smarter transportation management

11. Simple RL Architecture (Block Diagram)





12. Conclusion

Reinforcement Learning provides an intelligent solution for optimizing transportation systems such as bus scheduling and route allocation. By modeling the environment through **states** (traffic conditions, passenger demand, bus occupancy), selecting appropriate **actions** (route changes, bus dispatching, scheduling), and maximizing **rewards** (reduced delays, lower fuel consumption, improved passenger satisfaction), RL continuously improves transportation performance. Real-time data from GPS, traffic sensors, and passenger information enables the system to adapt dynamically to changing conditions, resulting in more efficient, reliable, and cost-effective public transportation.

4. Apply Reinforcement Learning to dynamic pricing in e-commerce. Define the state space, action space, and reward function. Discuss challenges in balancing exploration and exploitation.

1. Introduction

Dynamic pricing is a strategy where product prices are adjusted automatically based on market conditions, customer demand, competitor pricing, inventory levels, and buying behavior. Reinforcement Learning (RL) helps e-commerce platforms learn the optimal pricing strategy by continuously interacting with customers and maximizing long-term revenue and profit.

2. Why Reinforcement Learning?

Reinforcement Learning is suitable because it:

- Learns customer purchasing behavior over time.
- Adapts to changing market conditions.
- Optimizes long-term revenue rather than short-term sales.
- Automatically adjusts prices in real time.
- Improves customer satisfaction by offering competitive prices.

3. State Space

The **state space** represents the current business and market conditions that influence pricing decisions.

Example State Variables

- Current product price
- Customer demand
- Competitor prices
- Inventory level
- Time of day
- Season or festival period
- Customer purchase history
- Product popularity

Example State

State S =

```
{  
Product Price = ₹1200,  
Demand = High,  
Inventory = 150 units,  
Competitor Price = ₹1180,  
Weekend Sale  
}
```

4. Action Space

The **action space** consists of all possible pricing decisions the RL agent can make.

Examples include:

- Increase price by 5%
- Increase price by 10%
- Decrease price by 5%
- Decrease price by 10%
- Keep the current price unchanged
- Offer a promotional discount
- Apply bundle pricing
- Launch a limited-time offer

5. Reward Function

The reward function measures how successful a pricing decision is.

Positive Rewards (+)

Action	Reward
Product sold with high profit	+10
Increased revenue	+8
Higher customer purchases	+7
Inventory sold efficiently	+6
Customer returns for repeat purchase	+9

Negative Rewards (−)

Action	Reward
No product sold	−8
Low profit margin	−5
Excess inventory remains	−6
Customer abandons cart	−7
Price higher than competitors	−5

6. Exploration vs. Exploitation

One of the biggest challenges in RL is balancing **exploration** and **exploitation**.

Exploration

The agent tries new pricing strategies to discover whether they generate higher profits.

Examples:

- Testing a 10% discount.
- Trying premium pricing.
- Offering flash-sale prices.

Advantage: Finds better pricing strategies.

Disadvantage: May temporarily reduce profits.

Exploitation

The agent uses the pricing strategy that has already produced good results.

Examples:

- Keeping the price at ₹1200 because it has consistently generated high sales.
- Repeating successful promotional offers.

Advantage: Maximizes current revenue.

Disadvantage: May miss better pricing opportunities if market conditions change.

7. Challenges in Balancing Exploration and Exploitation

- Too much exploration can reduce profits by testing ineffective prices.
- Too much exploitation may prevent the system from discovering more profitable pricing strategies.
- Customer preferences change over time.
- Competitors frequently adjust their prices.
- Seasonal demand causes fluctuations in optimal pricing.
- Inventory levels continuously change.

An effective RL system balances both approaches to achieve maximum long-term profit.

8. Learning Process

The RL agent performs the following steps:

1. Observe the current market state.
2. Select a pricing action.
3. Display the new product price.
4. Monitor customer responses (purchase or no purchase).
5. Receive a reward based on revenue and profit.
6. Update its pricing policy.
7. Repeat the process continuously.

Over time, the agent learns which pricing strategies maximize long-term business performance.

9. Real-World Example

Amazon Dynamic Pricing

Amazon uses AI-based pricing systems to adjust product prices based on:

- Customer demand
- Competitor prices
- Inventory availability
- Seasonal sales
- Customer purchasing behavior

The pricing system continuously learns from customer interactions and updates prices automatically to maximize revenue while remaining competitive.

Other examples include:

- Flipkart
- Walmart
- eBay
- Alibaba

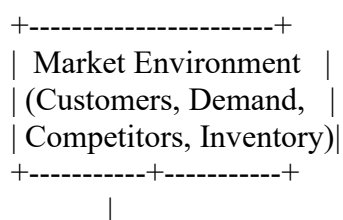
10. Advantages of Reinforcement Learning

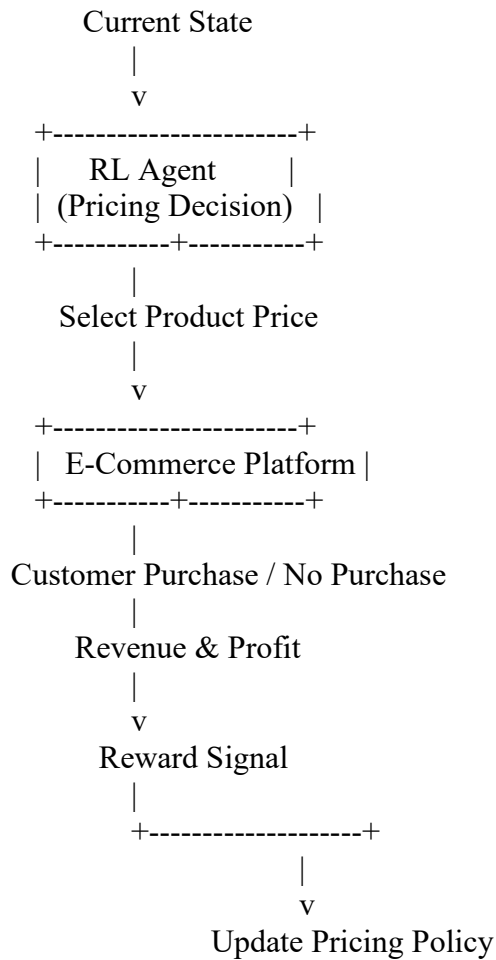
- Maximizes revenue and profit
- Adapts to market changes automatically
- Responds to customer demand in real time
- Optimizes inventory management
- Reduces manual pricing decisions
- Improves competitiveness
- Supports personalized pricing strategies
- Continuously improves pricing accuracy

11. Expected Outcomes

- Increased sales revenue
- Higher profit margins
- Better inventory utilization
- Improved customer satisfaction
- Competitive pricing
- Reduced unsold inventory
- Faster response to market changes
- Long-term business growth

12. Simple RL Architecture (Block Diagram)





13. Conclusion

Reinforcement Learning provides an intelligent solution for dynamic pricing in e-commerce by continuously learning from market conditions and customer behavior. It models the environment using **state space** (demand, inventory, competitor prices), selects pricing **actions** (increase, decrease, or maintain prices), and optimizes decisions through a **reward function** based on revenue and customer response. Although balancing **exploration** and **exploitation** is challenging, an effective RL system can maximize long-term profits while adapting to changing customer preferences and competitive market conditions.

5. Illustrate the use of Reinforcement Learning in smart city waste management systems. Identify the MDP components and discuss how continuous updates improve operational efficiency.

1. Introduction

Smart city waste management aims to collect and dispose of waste efficiently while minimizing operational costs and environmental impact. Reinforcement Learning (RL) helps optimize waste collection routes, collection schedules, and resource allocation by learning from real-time data such as bin fill levels, traffic conditions, and vehicle locations. The goal is to reduce fuel consumption, prevent overflowing bins, and improve the overall cleanliness of the city.

2. Why Reinforcement Learning?

Reinforcement Learning is suitable because it:

- Learns from real-time waste collection data.
- Adapts to changing waste generation patterns.
- Optimizes collection routes dynamically.
- Reduces fuel consumption and travel time.
- Improves resource utilization.
- Continuously improves operational efficiency.

3. Markov Decision Process (MDP) Components

a) Agent

The intelligent waste management system that decides which bins to collect and selects the best routes.

b) Environment

The smart city environment, including:

- Smart waste bins
- Collection trucks
- Road network
- Traffic conditions
- Waste generation patterns

c) State (S)

The current condition of the waste collection system.

Example State Variables

- Bin fill level
- Truck location
- Traffic congestion
- Fuel level
- Time of day
- Weather conditions
- Number of available trucks

Example State

State S =

```
{
Bin Fill = 90%,
Truck at Zone A,
Traffic = Moderate,
Fuel = 60%,
Time = 9:00 AM
}
```

d) Action (A)

The actions available to the RL agent.

Examples:

- Dispatch a collection truck
- Select the shortest route
- Select the least congested route
- Delay collection
- Prioritize overflowing bins
- Assign another vehicle
- Skip empty bins

e) Reward (R)

The reward measures the effectiveness of each decision.

Positive Rewards (+)

Action	Reward
Overflow prevented	+10
Short travel distance	+8
Reduced fuel consumption	+7
On-time waste collection	+9
Efficient route completed	+8

Negative Rewards (-)

Action	Reward
--------	--------

Action	Reward
Bin overflow	−10
Long travel distance	−6
Traffic delay	−5
High fuel consumption	−7
Missed collection	−8

f) Policy (π)

The policy determines the best action for each state.

Example:

- If a bin is almost full and traffic is low → Dispatch the nearest truck.
- If traffic is heavy → Choose an alternative route.
- If a bin is nearly empty → Delay collection.

4. Learning Process

The RL agent follows these steps:

1. Observe the current state.
2. Select the best collection action.
3. Execute the waste collection plan.
4. Receive a reward based on efficiency.
5. Update the policy using feedback.
6. Repeat the process continuously.

Over time, the system learns the most efficient waste collection strategies.

5. Continuous Updates Improve Operational Efficiency

The RL system continuously receives real-time information from:

- Smart bin sensors
- GPS-enabled collection trucks
- Traffic monitoring systems
- Weather forecasts
- City surveillance systems

Benefits of Continuous Updates

- Dynamic route optimization
- Faster response to overflowing bins
- Reduced fuel consumption
- Lower operational costs
- Better truck utilization

- Improved city cleanliness
- Reduced carbon emissions
- Higher service reliability

As new data arrives, the RL agent updates its policy and makes better decisions without manual intervention.

6. Real-World Example

Smart Waste Collection System

A smart city installs IoT sensors in waste bins to monitor fill levels.

- Sensors detect when bins are nearly full.
- GPS tracks waste collection trucks.
- Traffic data identifies the fastest routes.
- The RL agent decides which bins to collect first and assigns the nearest available truck.
- The system continuously updates routes based on changing traffic and waste levels.

This reduces unnecessary trips, saves fuel, and ensures timely waste collection.

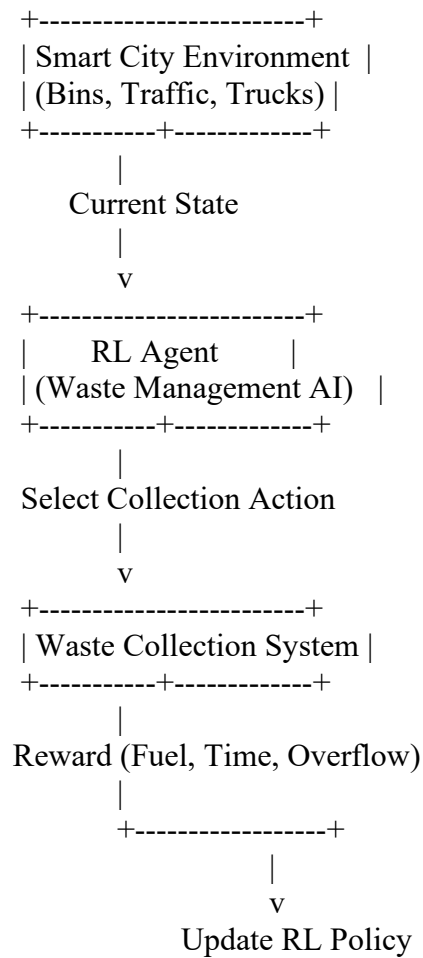
7. Advantages of Reinforcement Learning

- Optimizes waste collection routes
- Reduces operational costs
- Prevents overflowing bins
- Minimizes fuel consumption
- Improves resource utilization
- Adapts to changing city conditions
- Supports real-time decision making
- Enhances environmental sustainability
- Continuously improves through experience

8. Expected Outcomes

- Cleaner urban environment
- Reduced fuel and maintenance costs
- Faster waste collection
- Improved collection efficiency
- Lower greenhouse gas emissions
- Better utilization of waste collection vehicles
- Increased citizen satisfaction
- Smarter and more sustainable waste management

9. Simple RL Architecture (Block Diagram)



10. Conclusion

Reinforcement Learning enables smart city waste management systems to optimize collection routes, schedules, and resource allocation by modeling the problem as a **Markov Decision Process (MDP)** with clearly defined **states, actions, rewards, and policies**. By continuously learning from IoT sensors, GPS data, and traffic information, the RL agent adapts to changing conditions and improves operational efficiency over time. This leads to reduced costs, lower fuel consumption, cleaner cities, and more sustainable waste management practices.

Code of Conduct:

*I Malli Chandana(192472250)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Malli chandana

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation